Permian Basin Load Interconnection Study - Status Update

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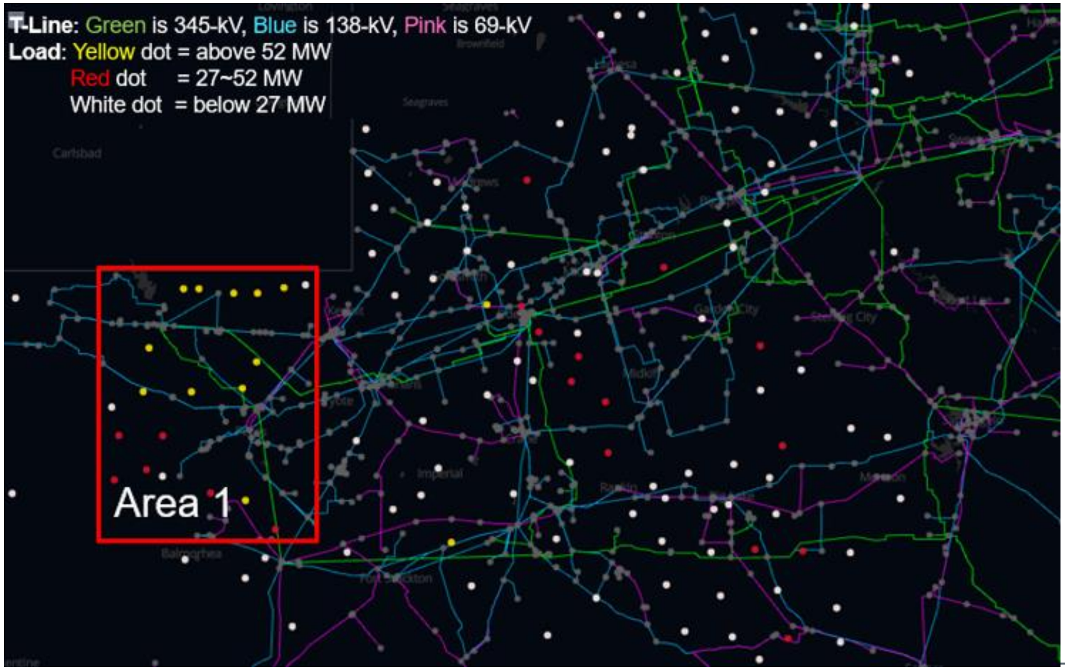
Recap

- Purpose of the Study
  - Identify local transmission upgrades that may be necessary to connect potential local oil and gas loads in the Permian Basin area
- ERCOT provided status updates at the previous RPG meetings
  - Permian Basin Area Improvement Updates at the December 2020 RPG meeting http://www.ercot.com/content/wcm/key_documents_lists/189757/Permian_Basin_Area_Improvement_Updates_-_Dec2020RPG.PDF
  - Permian Basin Load Interconnection Study Scope at the January 2021 RPG meeting http://www.ercot.com/content/wcm/key_documents_lists/213837/Permian_Basin_Load_Interconnection_Study_Scope__-_Jan2021RPG.pdf
  - Permian Basin Load Interconnection Study Status Update at the April 2021 RPG meeting http://www.ercot.com/content/wcm/key_documents_lists/213851/Permian_Basin_Load_Interconnection_Status_Update__-_April2021RPG.pdf
  - Permian Basin Load Interconnection Study Status Update at the June 2021 RPG meeting http://www.ercot.com/content/wcm/key_documents_lists/213859/Permian_Basin_Load_Interconnection_Status_Update__-_June2021RPG.pdf

Disclaimer

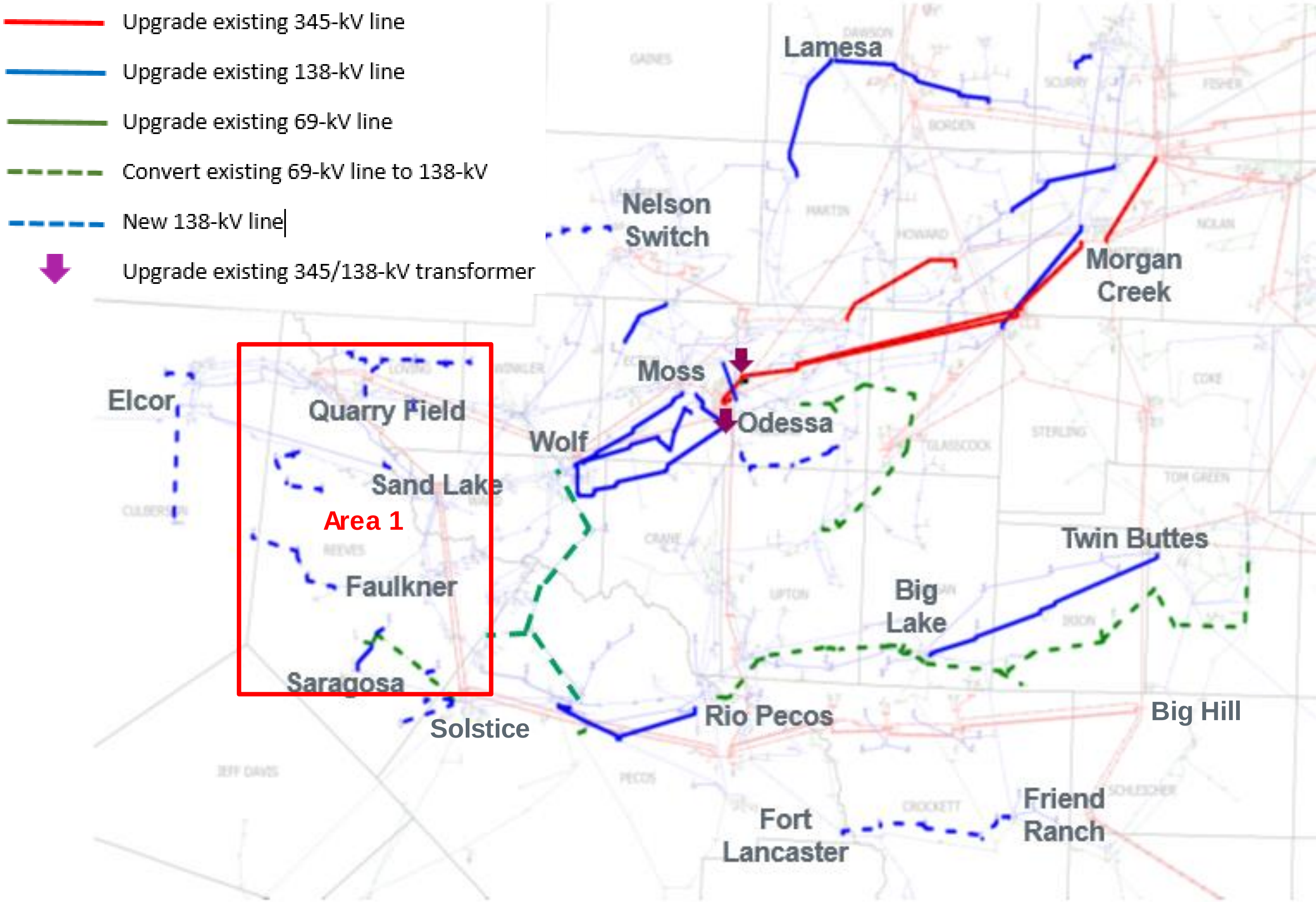
- For the new loads, this study focuses on Area 1 in Delaware Basin for the transmission interconnection based on the following considerations:
  - ➢ Size and concentration of new loads
  - ➢ Lack of existing transmission infrastructure
  - ➢ Location profitability
- For the new loads outside of Area 1, additional RPG review may be required for the transmission interconnection study



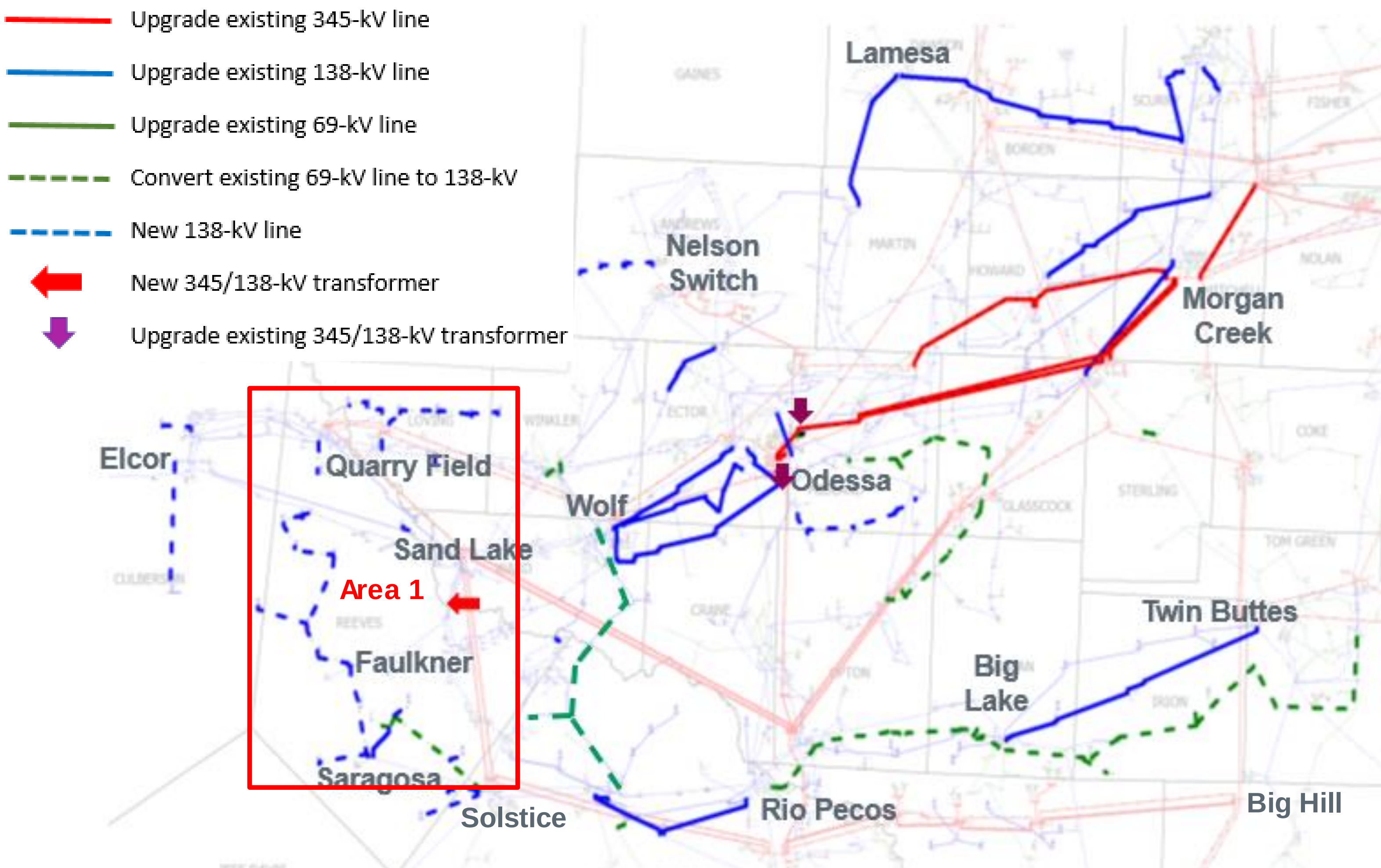
Status Update

- Preliminary reliability upgrades have been identified for the 2025 and 2030 cases
  - New load connections and upgrades
  - Certain existing transmission facility overload issue
- The relevant TSPs are currently reviewing the preliminary reliability upgrades and estimating the cost of each upgrade

Preliminary Reliability Upgrades - 2025



Preliminary Reliability Upgrades - 2030



Stage 2 Upgrade (Bearkat – North McCamey – Sand Lake 345-kV double ckt) is applied to 2030 case as a placeholder project

Next Step and Tentative Timeline

- ERCOT and relevant TSPs will continue to work together to review the preliminary upgrades and cost estimates

- Complete the study in Q3 or Q4 2021

?

Stakeholder Comments Also Welcomed to Sun Wook Kang:

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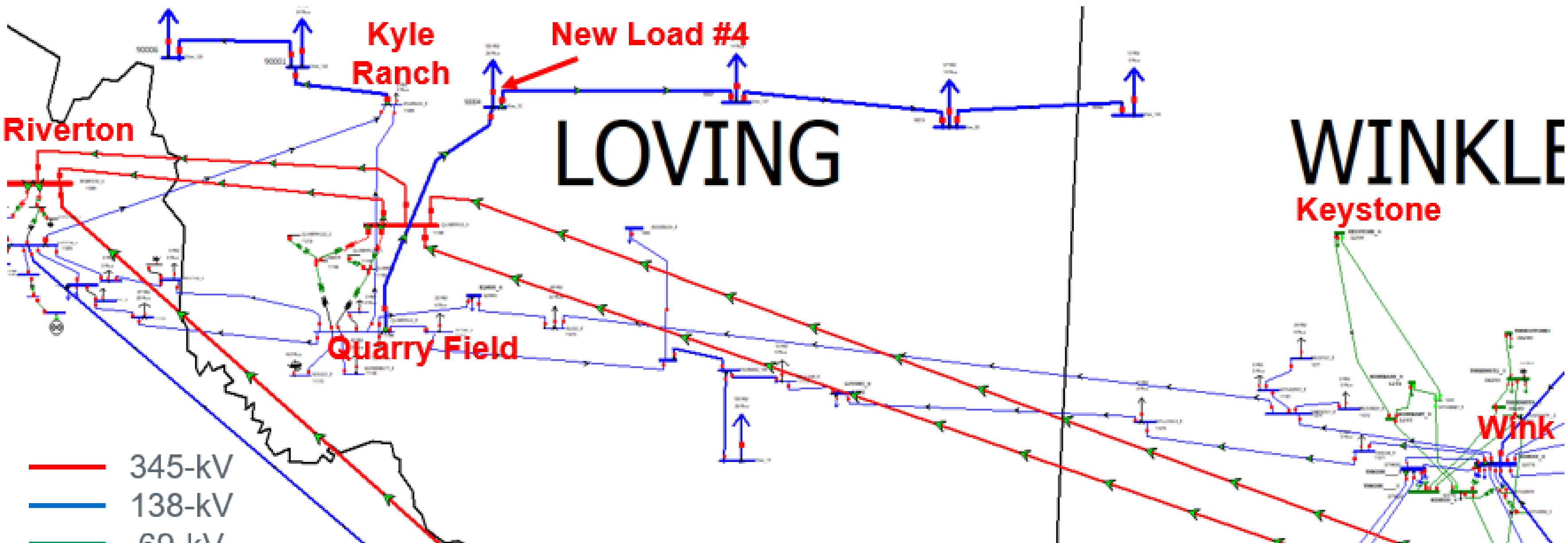
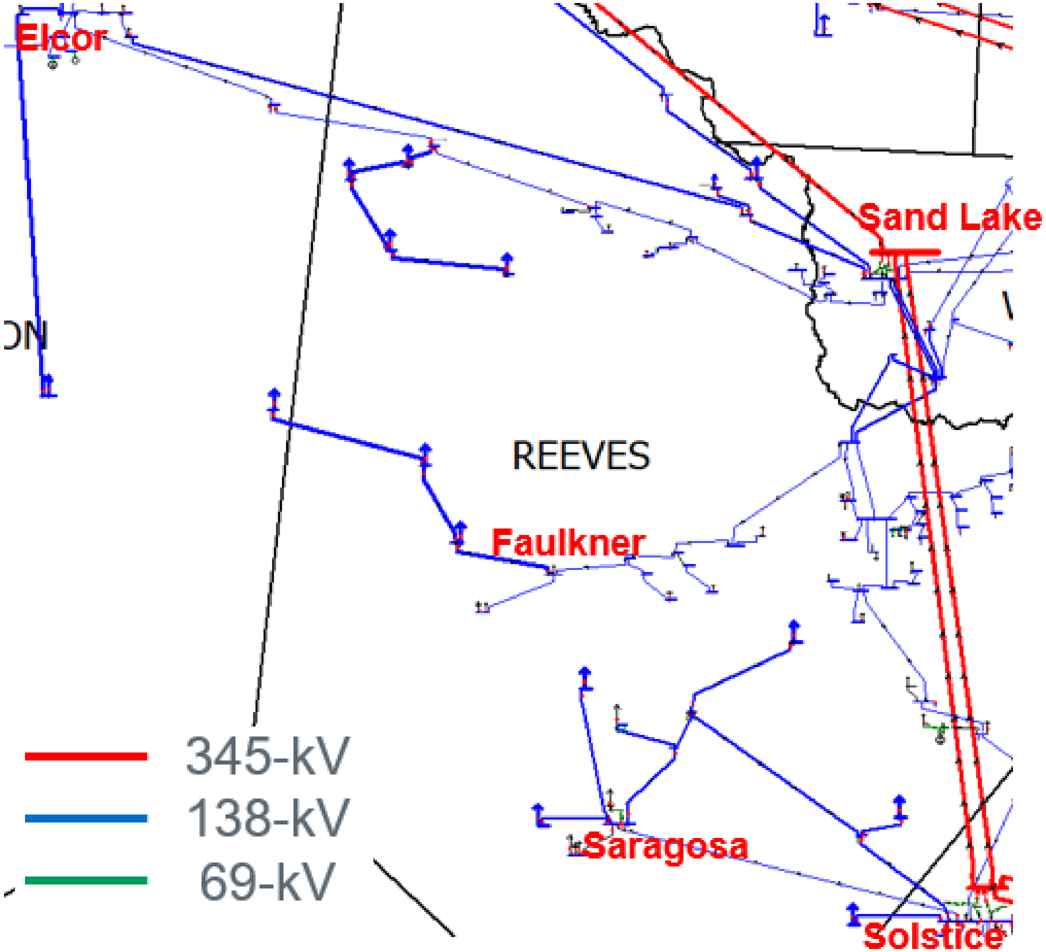
Appendix - New Load Connections at Culberson Loop

- No violations were found by radially connecting the four big loads (total 233 MW in 2030) to the Quarry Field substations. TSPs are reviewing the connections
- Tested the alternative of connecting New Load #4 to Kyle Ranch (~ 4 miles) instead of Quarry Field (~ 10miles). This will have more impact on the loading of Riverton transformer (Riverton auto 2 is overloaded under X1+N1 condition in this option)



For illustrative purpose

Appendix - New Load Connections in Reeves County

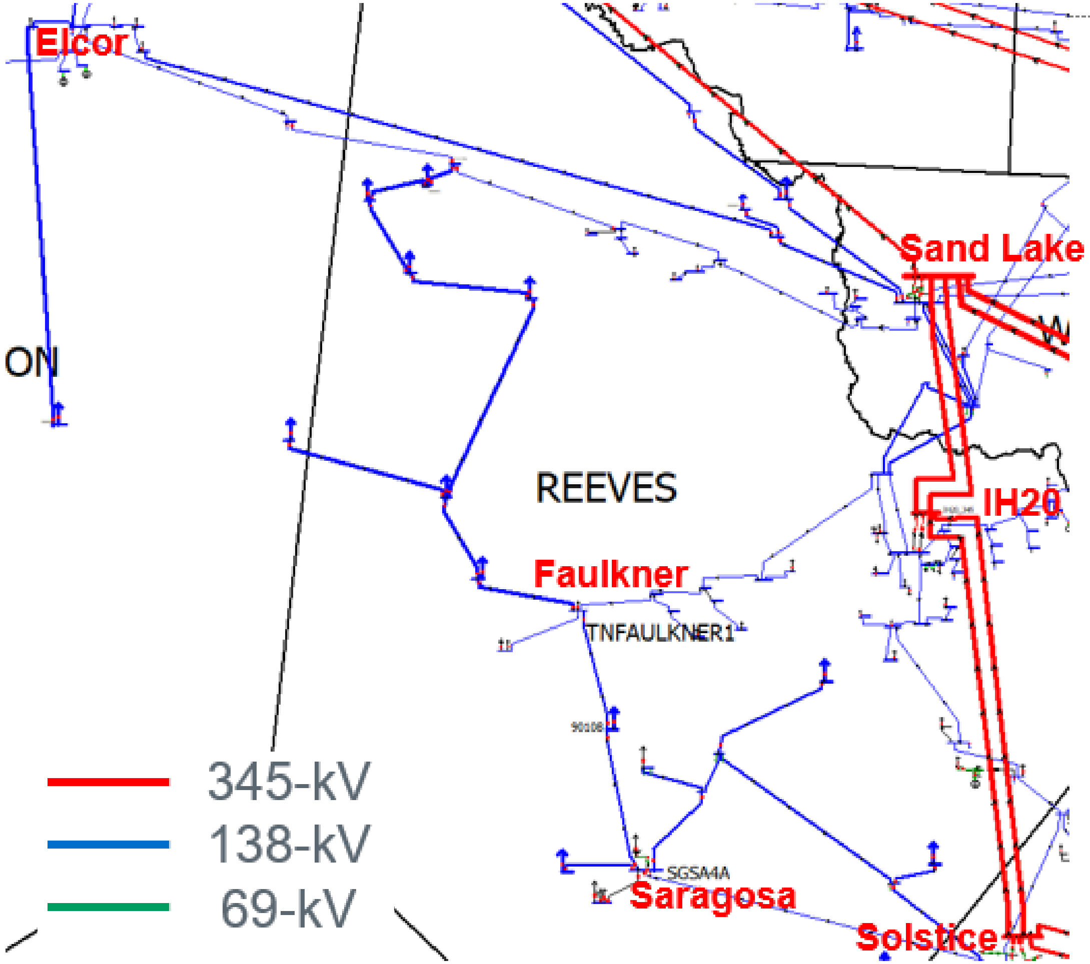


2025 Case:

- No violations were found by radially connecting the new loads in 2025 case (Note: reactive devices were assumed at the new load buses to address the reactive needs)
- TSPs are reviewing the new load interconnections

For illustrative purpose

Appendix - New Load Connections in Reeves County (Cont.)



For illustrative purpose

2030 Case:

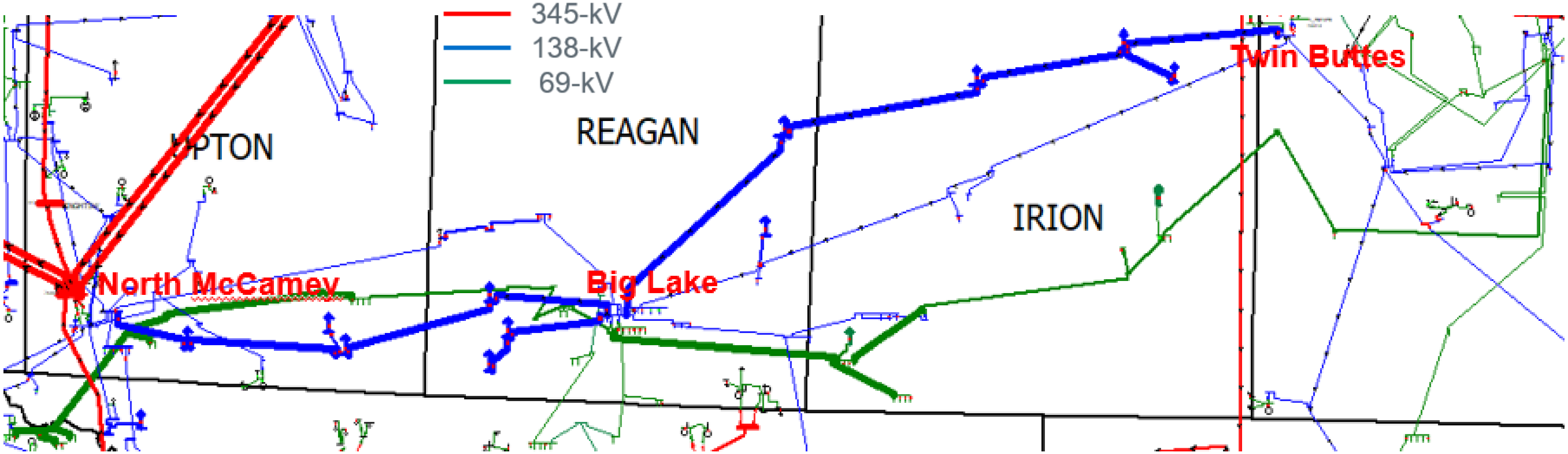
- Need to Loop-In-Loop-Out of Solstice – Sand Lake 345-kV double circuits near IH20 substitution and add two new 345/138-kV transformers
- Need connections between Faulkner and Saragosa (Note: reactive devices were assumed at the new load buses to address the reactive needs)

Appendix - New 345/138-kV Transformation Near IH20

- The existing Solstice – Sand Lake 345-kV line needs to be looped into the 138-kV system near IH-20 Substation in order to address reliability issues
- Three Options Evaluated
  - ➢ Loop-In-Loop-Out at IH20 Substation
    - ✓ Physical space limitation
    - ✓ No further upgrades required
  - ➢ Loop-In-Loop-Out at Collie Field Substation
    - ✓ Need to upgrade 2.95 miles of existing 138-kV line from Collie Field Tap to IH20
  - ➢ Loop-In-Loop-Out at Saddleback Substation
    - ✓ Need to upgrade 4.88 miles of existing 138-kV line from Saddleback to IH20

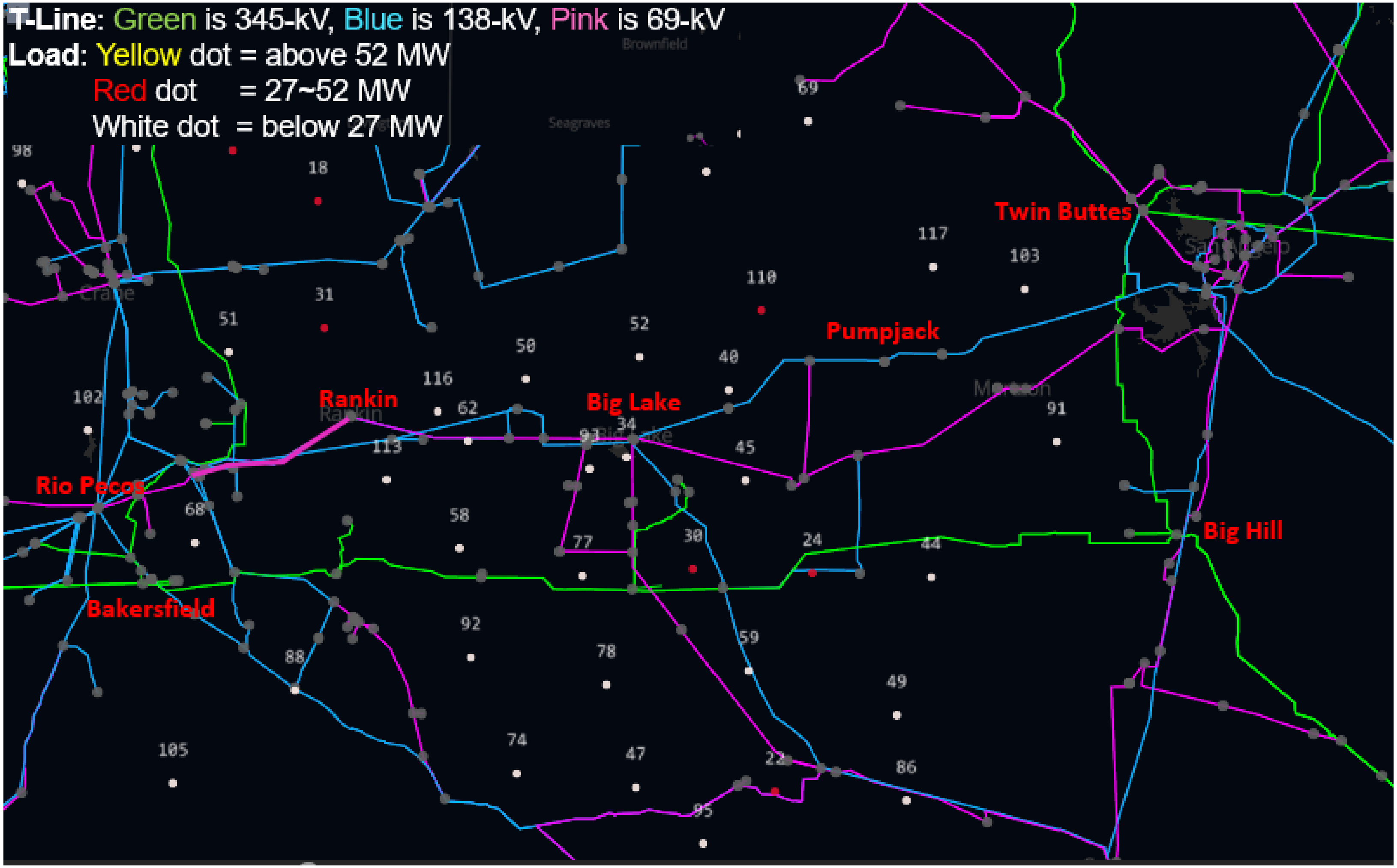
Appendix - New Load Connections at Big Lake Area

- Thick blue lines are new 138-kV lines added by TSP to connect the new loads at Big Lake area
- Thick green lines are the existing 69-kV lines which need to be converted to 138-kV even with the new 138-kV lines (with the new load connections from TSP)
- The existing 138-kV lines from Big Lake to Twin Buttes ~~needs~~ need to be upgraded
- Since this area involves new transmission as well as upgrading existing transmission, further studies were performed to evaluate the system needs



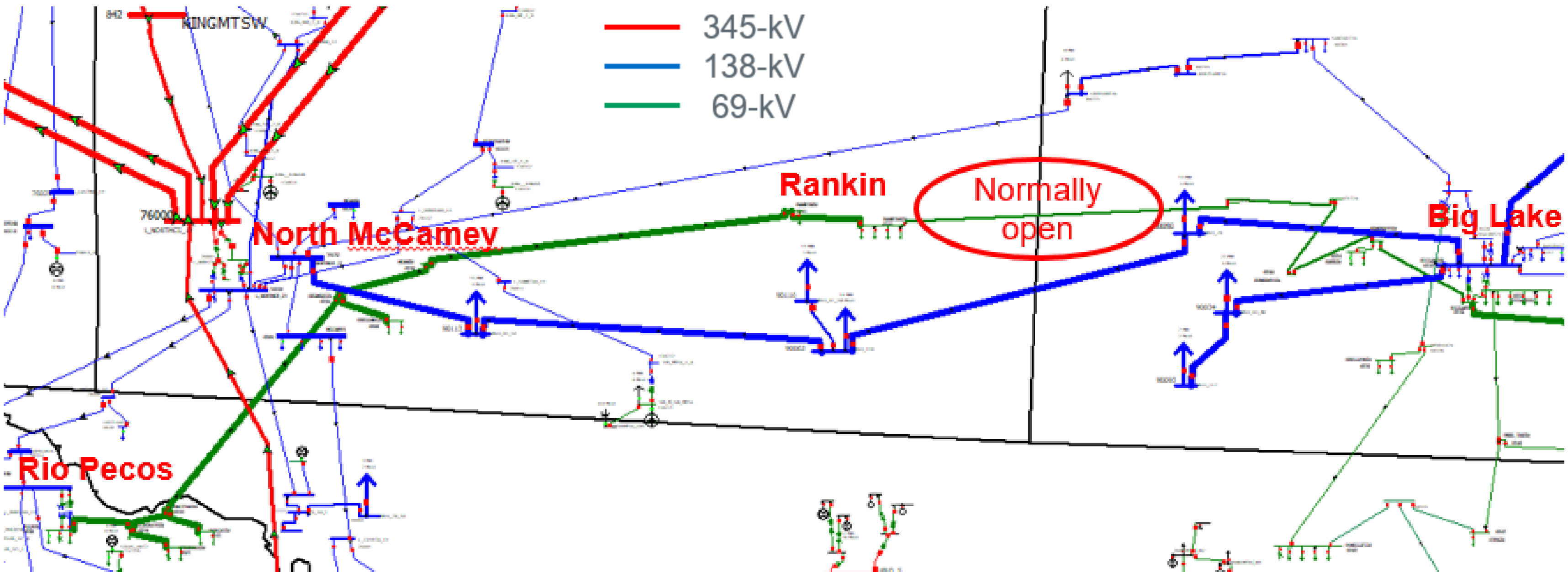
For illustrative purpose

Appendix - Map of the new loads at Rio Pecos to Big Lake to Twin Buttes



Appendix - New Load Connections at North McCamey to Big Lake Area

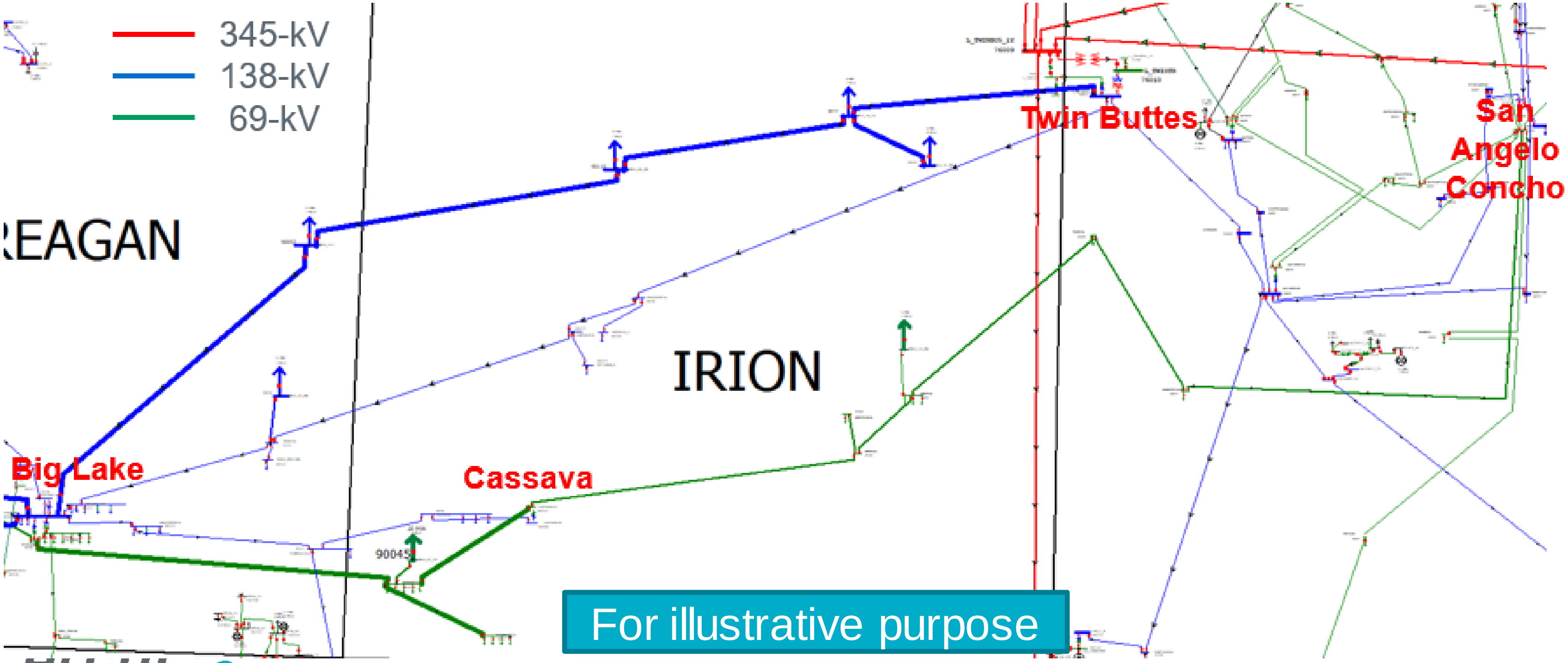
- The new loads (~ 55 MW in 2030) along North McCamey to Big Lake are close to the existing transmission lines
- The existing 69-kV lines from Rio Pecos to Rankin needs to be converted to 138-kV even with the new 138-kV line
- Converting the 69-kV line from Rio Pecos all the way to Big Lake to 138-kV and radially connect the new loads to the existing substations were identified as placeholder projects



For illustrative purpose

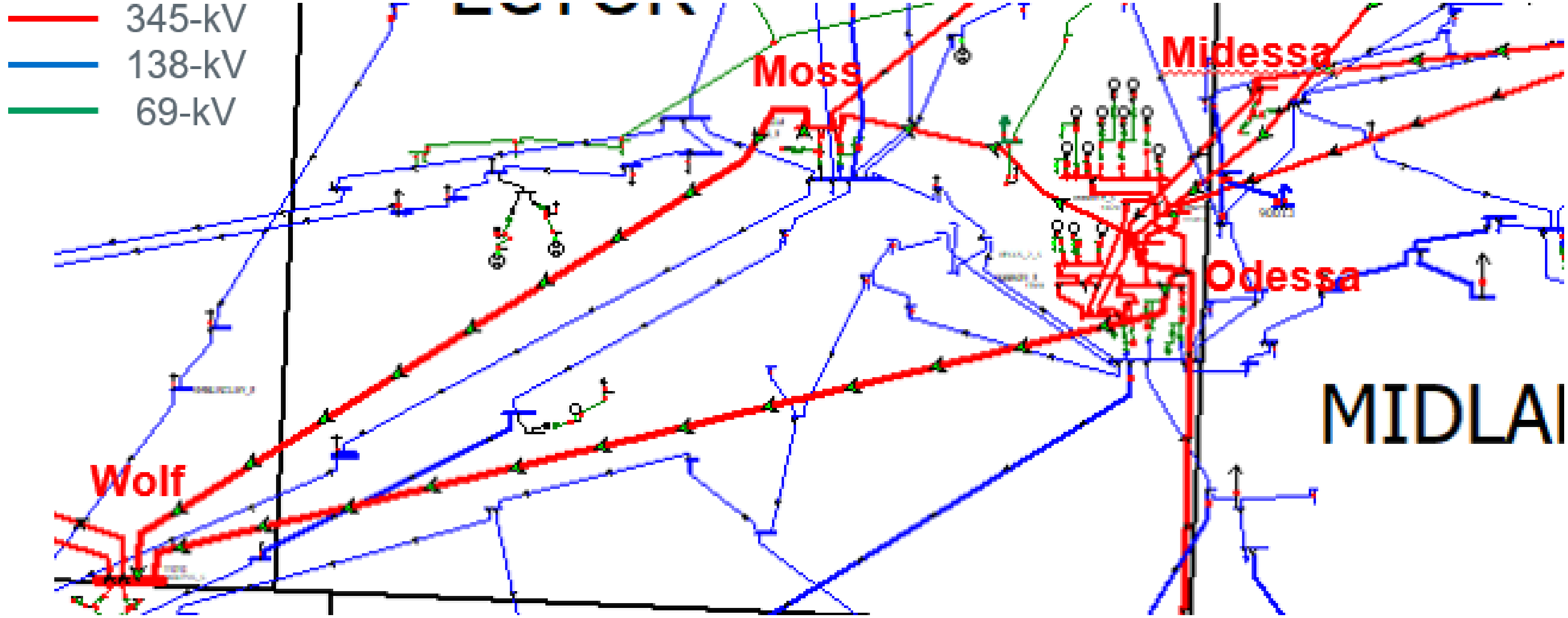
Appendix - New Load Connections at Big Lake to Twin Buttes Area

- With the new loads (~ 56 MW in 2030) in Big Lake to Twin Buttes area and transmission connections provided by TSP, additional upgrades are required
  - Upgrade the existing 138-kV line from Big Lake to Twin Buttes
  - Convert the portion of the 69-kV line from Big Lake to Cassava to 138-kV (due to the new loads)
- Radially connecting the new loads to the existing 138-kV line and converting the 69-kV line from Big Lake all the way to San Angelo Concho to 138-kV were identified as a placeholder project



Appendix - Odessa #2 Transformer Overload – One Example

- Odessa transformer #2 is heavily overloaded under the X1+N1 condition (outage of Odessa transformer #1 plus Odessa-Moss and Odessa-Wolf P7)
- The ratings for Odessa transformers #1 and #3 are 470/544 MVA, and 627/706 MVA for transformer #2 (however, transformer #2 has lower reactance)



For illustrative purpose

Appendix - Options to Address Odessa #2 Transformer Overload

Options	2025 Case (%)	2030 Case (%)
2nd Midessa auto	121.3	128.7
2nd Midessa auto + 2nd Newswitch auto + Add IH20 345-kV station and autos in 2025	117.6	125.5
Upgrade the three Odessa autos (700/750 MVA)	98.5	98.0
Add new Midessa to Moss 345-kV line (~ 20 miles)	96.1	98.1
Tap a new 345-kV substation on the Odessa - King Mountain 345-kV line (two miles away from Odessa) and add one auto	106.6	107.4
Upgrade the three Odessa autos (700/750 MVA) + Tap a new 345-kV substation on the Odessa - King Mountain 345-kV line (two miles away from Odessa) and add one auto	77.1	78.7
2nd Midessa auto + Tap North McCamey - Sand Lake 345-kV ckt 1 at Riverview and add one auto	n/a	125.7
2nd Midessa auto + New 138-kV line from Midessa to Rexall (~ 10 miles)	n/a	123.4
2nd Midessa auto + 2nd 138-kV circuit from Midessa to Odessa (~ 11 miles)	n/a	124.8

Appendix - Options to Address Odessa #2 Transformer Overload (Cont.)

- Tap a New 345-kV Substation on the Odessa - King Mountain 345-kV Line